

Volve 15/9-F-12B Geological & Drilling Hazards Report

Section: Stratigraphy & Overview

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Well 15/9-F-12B was spudded as a production well targeting the Hugin and Heimdal sandstone reservoir intervals. The primary geomechanical objective was reaching 3450m MD through complex interbedded shale and permeable sands.

Subsurface stratigraphic horizons include Hordaland shale from 1800m to 2900m, Balder reactive smectite formation at 3100m, and Heimdal sandstone formation between 3230m and 3270m MD.

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Section: Extraction Failure Notes & Sandstone Geohazards

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Extraction failure notes in the adjacent sandstone layers indicate severe differential sticking at 3245m MD. The active BHA became immobilized after a 15-minute connection due to 450 psi differential overbalance.

Formation evaluation revealed highly permeable sandstone (porosity 28%,) Tj ET BT /F1 10 Tf 50 606 Td (permeability 850 mD)

overbalance caused mechanical binding of the drill collars against the borehole wall.

Immediate remediation required continuous jarring for 36 hours and pumping 50 bbl of pipe-freeing lubricating chemical wash. Total non-productive time (NPT) was 48.0 hours with remediation cost exceeding \$420,000.

Mitigation recommendations for future offset drilling: Maintain rotary speed at 60 RPM, limit stationary time to under 3 minutes, maintain WOB below 14 klbs, and optimize fluid loss additives to ensure a ultra-thin, low-friction mud filter cake.

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Section: Carbonate Fractures & Severe Mud Losses

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At 3290m MD within the Lower Hordaland fractured carbonate boundary, sudden partial to total mud losses occurred at a rate exceeding 350 bbl/hr.

Hydrostatic head decreased by 180 psi, prompting immediate deployment of 75 bbl coarse and medium LCM pills. Pump flow rate was throttled to 450 GPM to reduce Equivalent Circulating Density (ECD) until losses subsided.

Recommended mitigation: Increase mud weight by 0.2 ppg incrementally once fractures seal to ensure borehole mechanical stability while monitoring PWD sensors.